Reg. No. | | | | | | | | | | | [12EE55/13EE55]

FIFTH SEMESTER B.E/B.TECH EXAMINATION, DECEMBER–2015

OPAMPS AND LIC

Time:3 Hours **Maximum Marks:70**

Instructions:

i. Answer any **FIVE** full Questions.

ii. Use of Data sheet is permitted.

1. a. Define the following for an OPAMP: **08**

i. Virtual ground

ii. CMRR

iii. PSRR

iv. Input offset current

b. Discuss the ideal voltage transfer curve of an OPAMP. **03**

c. Explain a voltage follower circuit along with a connection diagram. **03**

2. a. Draw the neat circuit of the high z_{in} capacitor coupled non-inverting amplifier and derive the expression for input impedance. **08**

b. Design a capacitor coupled inverting amplifier to have cut off frequency f_1=100hz, f_2=10khz. The input impedance should be 5.6 KΩ and the load impedance is 470Ω, voltage gain = 5. **06**

3. a. What is frequency compensation? Explain the phase lag compensation and show how it affects the frequency response of an operational amplifier. **07**

b. Define slew rate of the opamp. Calculate the slew-rate limited cut-off frequency for a voltage follower using opamp 741 if the maximum amplitude of a sine wave is 6V. Also find out the maximum amplitude of the sine wave output if the unity gain cut off frequency is 800khz. Consider slew rate s=0.5 V/ μ sec. **03**

c. Explain the minimization of stray capacitance effect in OPAMP. **04**

4. a. With a neat circuit diagram and wave forms explain the operation of OPAMP sample and hold circuit. **09**

b. A ±5V, 10 khz square wave from a signal source with a resistance of 200Ω has its positive peak clamped precisely at ground level. Tilt on the output is not to exceed 1% of the peak amplitude of the wave. Design a precision clamping circuit using a supply of ±15V. **05**

ENGINEERING & TECHNOLOGY LIBRARY DATE ..... VTU UNIVERSITY-BANGALORE

5. a. Draw the circuit of an OPAMP monostable multi vibrator, with the voltage waveforms at different points and explain its operation. Also show the design aspects. **10**

b. Using 741 OPAMP, design a voltage level detector circuit to switch its output between approximately -17 V and +17 V, when the input exceeds 1 V. **04**

6. a. With a neat diagram and relevant waveforms explain the operation of a triangular/rectangular waveform generator which has frequency and duty cycle controls. **10**

b. Using a BIFET OPAMP with a ±15V power supply, design a Wein bridge oscillator to have an output frequency of 12khz. **04**

7. a. Sketch the circuit of a second order active high pass filter. Explain its operation with the expected frequency response. **07**

b. Design a second order high-pass active filter to have a cut-off frequency of 12khz. Use a 715 OPAMP and estimate the highest signal frequency that will be passed (For 715 OPAMP, I_B(max) = 1.5 μA and fu=11 MHz) **07**

8. a. Design a OPAMP series voltage OPAMP regulator to meet the following specification: **09**

V_i=(18 ± 3) volts (input voltage)

V_0=9 volts at I_0=10 to 50 mA $\begin{pmatrix} V_\circ = output\ voltage \\ I_\circ = output\ current \end{pmatrix}$

Zener diode voltage rating V_z=5.6V

Zener diode power dissipation (maximum) P_{zmax} = 0.5w

(assume minimum Zener current = 5 mA, β of transistor = 30, opamp 741 has output current rating = 25 mA]

b. Sketch the basic circuit of a 723 integrated circuit DC voltage regulator and briefly explain it. **05**